

1. Face verification requires comparing a new picture against one person's face, whereas face recognition requires comparing a new picture against K persons' faces. 1 point

☐ False

☒ True
2. Why do we learn a function $d(img1, img2)$ for face verification? (Select all that apply.) 1 point

☒ This allows us to learn to recognize a new person given just a single image of that person.

☐ This allows us to learn to predict a person's identity using a softmax output unit, where the number of classes equals the number of persons in the database plus 1 (for the final "not in database" class).

☒ We need to solve a one-shot learning problem.

☐ Given how few images we have per person, we need to apply transfer learning.
3. In order to train the parameters of a face recognition system, it would be reasonable to use a training set comprising 100,000 pictures of 100,000 different persons. 1 point

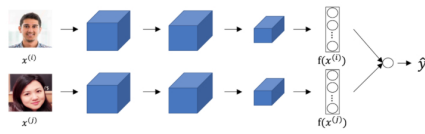
☒ True

☐ False
4. Which of the following is a correct definition of the triplet loss? Consider that $\alpha > 0$. (We encourage you to figure out the answer from first principles, rather than just refer to the lecture.) 1 point

☐ $\max(\|f(A) - f(N)\|^2 - \|f(A) - f(P)\|^2 + \alpha, 0)$

☐ $\max(\|f(A) - f(P)\|^2 - \|f(A) - f(N)\|^2 - \alpha, 0)$

☐ $\max(\|f(A) - f(N)\|^2 - \|f(A) - f(P)\|^2 - \alpha, 0)$

☒ $\max(\|f(A) - f(P)\|^2 - \|f(A) - f(N)\|^2 + \alpha, 0)$
5. Consider the following Siamese network architecture: 1 point


The upper and lower networks share parameters to have a consistent encoding for both images. True/False?

☒ True

☐ False
6. Our intuition about the layers of a neural network tells us that units that respond more to complex features are more likely to be in deeper layers. True/False? 1 point

☒ True

☐ False
7. Neural style transfer uses images Content C, Style S. The loss function used to generate image G is composed of which of the following: (Choose all that apply.) 1 point

☒ $J_{content}$ that compares C and G .

☐ T that calculates the triplet loss between S , G , and C .

☒ J_{style} that compares S and G .

☐ J_{corr} that compares C and S .
8. In neural style transfer, we define style as: 1 point

☐ The correlation between the activation of the content image C and the style image S .

☒ The correlation between activations across channels of an image.

☐ $\|a^{[l](S)} - a^{[l](C)}\|^2$ the distance between the activation of the style image and the content image.

☐ The correlation between the generated image G and the style image S .
9. In neural style transfer, which of the following better express the gradients used? 1 point

☒ $\frac{\partial J}{\partial G}$

☐ $\frac{\partial J}{\partial W}$

☐ $\frac{\partial J}{\partial S}$

☐ Neural style transfer doesn't use gradient descent since there are no trainable parameters.
10. You are working with 3D data. The input "image" has size $64 \times 64 \times 3$, if you apply a convolutional layer with 16 filters of size $4 \times 4 \times 4$, zero padding and stride 2. What is the size of the output volume? 1 point

☐ $64 \times 64 \times 3$.

☐ $31 \times 31 \times 31 \times 3$.

☐ $61 \times 61 \times 61 \times 14$.

☒ $31 \times 31 \times 31 \times 16$.